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**BIT 2324 GEOGRAPHICAL INFORMATION  
SYSTEMS**

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to [elarning@jkuat.ac.ke](mailto:elarning@jkuat.ac.ke)



## LESSON 6

### Spatial Data Files

#### Learning outcomes

By the end of this topic you should be able to;

- *Explain spatial data files*
- *Describe the presentation of the data files*



## 6.1. Introduction

Spatial data files are like other files you work with on a computer.

## 6.2. Characteristics of spatial data files

1. Spatial data files can be:

- stored on a hard drive, memory stick, CD, DVD
- assigned either a user-defined file name, or are given default file name by a software application
- organized into folders
- have the ability to be opened, viewed and edited by one or more GIS software applications that understand their format.

2. However, that is where the similarities end. Spatial data



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files are unique in that they store “georeferenced” information – information that defines location or place. In addition, descriptive information about the georeferenced information is stored in each spatial data file.

3. Thus rather than just text or numbers, an individual spatial data file is a digital representation of a similar group of geographic features on the surface of the earth.
4. The geographic features can be actual physical entities or events, or they can represent conceptual features.
5. Examples of individual spatial data files representing real geographic features or events are lakes, rivers, wetlands, elevation contours, roads, forested areas, rare species habitats, soils, earthquakes, vehicle thefts, electricity distribution lines, and groundwater reservoirs.



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6. Examples of individual spatial data files representing conceptual geographic features are census tract boundaries, zoning boundaries, or parcel boundaries i.e. conceptual features do not physically exist on the landscape, but are imposed by us for various reasons and can be represented in a geographic context.
7. Each spatial data file is uniquely constructed to work within GIS software applications and consists of unique characteristics:
  - “shapes” that attempt to reflect / convey the appearance and position of individual geographic features as accurately as possible
  - records within a related tabular database that contain numeric and/or textual descriptions of each feature



- a coordinate system that defines the true location of all the features on the earth's surface (i.e. the latitude/longitude)

### 6.3. Formats of Spatial Data Files

Spatial data files come in several different formats. You may need to use only one, or you may need to use a combination of them, depending on your particular application and/or type of analysis. Each format falls under one of two different categories: vector or raster.

- Vector spatial data files are ones in which the geographic features being represented are built by a collection of vertices and lines.
- Raster spatial data files are ones in which the geographic



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features across an entire area are represented by a continuous set of “pixels” or “cells”.

### 6.3.1. The Shapefile spatial data file format

This is a very common format for spatial data files in the vector category. In this format, geographic features can be represented in one of three ways:

- points
- lines
- polygons

If you utilize an existing shapefile, the point, line or polygon representation is chosen by the organization responsible for its development. If you create your own shapefile, you will have





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to determine which representation is best for your work or application. The determination is made based on several factors, including but not limited to:

- the need to depict features at a specific scale
- the need to depict features as realistically as possible
- the need to quantify some aspect of geographic features.

### 6.3.2. The Coverage spatial data file format

This was the original spatial data file format used in GIS software. While this format has taken a “backseat” to the shapefile format due to the simplicity of shapefiles, coverages are still very viable and have many advantages. Along with shapefiles, the coverage format is being superseded by the new geodatabase format. However, many web sites still offer spatial data for down-



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load in the coverage format. Just like shapefiles, geographic features are represented as points, lines, or polygons and many factors come into play when deciding which representation is best. Coverages also fall within the vector category. Unlike shapefiles, a single coverage is actually comprised of two folders. Each folder contains a multitude of other files that the GIS software “puts together” in order to represent geographic features and associated tabular information when it is opened. If either folder is missing, or if files from within either folder are missing, the coverage will be “corrupt” and not useable.

### 6.3.3. The Grid spatial data file format

In most respects, grids are very different from either shapefiles or coverages. Grids fall into the raster category; they are con-



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structured of rows and columns of pixels instead of vertices and arcs. Like coverages however, grids are comprised of two folders, each containing files that the software “puts together” for display and manipulation. Grids can be either:

- Integer Grids: The pixel values are integers and each integer may also be associated with one or more textual descriptions.
- Floating Point Grids: The pixel values will be expressed as decimals. Floating point grids can not have textual descriptions.

### 6.3.4. Images as Spatial Data Files

Many different image formats can be used in GIS. Some of these may be more familiar than others: .jpg, .tif, .bil, .png, .img,



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.sid. All image formats fall within the raster category of spatial data. In some cases, images are not used specifically as “spatial data”, but are used to enhance spatial data by providing a digital photograph of a place or object.

In other cases, the images themselves are spatial data. Data provided from the Landsat satellite is an example of imagery that is spatial. If you have ever used Google Earth, the images that appear when you zoom in are spatial data. When an image is “georeferenced” - meaning that information is embedded within the image that describes its position on the surface of the earth in real world coordinates (latitude/longitude), it becomes spatial data. In addition to being “georeferenced”, many images may also be “orthorectified”. This term refers to a complex process wherein distortions caused by differences in terrain



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elevation, camera tilt, and edge effects are removed from the image. Images that are both georeferenced and orthorectified are frequently called “orthophotographs” or just “orthos” in short.

### 6.3.5. Computer-Aided Drafting (CAD) files

CAD software applications such as AutoCAD and MicroStation produce vector data in .dwg or .dxf format. ArcGIS software can read these files directly, they do not need to be imported or changed in any way. In most instances, but not all, these .dwg or .dxf files will include georeferencing information and thus, the data are spatial. For those files that are not georeferenced, there are methods and / or separate software programs that allow them to become georeferenced. CAD files often provide very detailed information such as the floor plan of a building



which can be useful for certain GIS applications.

### 6.3.6. The GeoDataBase (GDB)

The GDB is the newest spatial data file format developed by ESRI. It is a replacement file format for all of the above. GDBs have very specific advantages over the other file types:

1. All spatial data resides within a single file. This eliminates having several different file formats residing in several different folders across a disk. The file has an .mdb extension i.e. a GDB containing all spatial data for Nairobi might be named nairobi.mdb
2. GDBs have greater “intelligence”
3. Industry specific data models can be applied to a GDB



## 6.4. Naming conventions for Spatial Data Files & Folders containing Spatial Data Files

Since spatial data files have a complex structure, it is extremely important that you follow some basic rules for assigning names to these files as well as to the folders that contain these files. Thus, to avoid these problems from ever happening to you, follow these simple rules:

- Keep the names short
  - Between 8 and 15 characters
  - If you work with spatial data that are in the “coverage” vector format, or the “grid” raster format, only 8 characters are allowed.
- Do Not Use Spaces



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- town boundaries is a bad name for a file; use town\_boundaries instead
- my gis data is a bad name for a folder that will contain spatial data files
- Do Not Use Special Characters (the only exception is the underscore)
  - e.g. town#boundaries is a bad name for a file
  - e.g. my!gis!data is a bad name for a folder
- Never Put A Number As The First Character
  - e.g. 2000population is a bad name for a file
  - a good alternative would be y2000pop
- Keep The Names Simple
  - use abbreviations where possible





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- use underscores where possible as a replacement for spaces
- use upper case where it makes sense
- e.g. twn\_bndy or TwnBndy
- Implement A Tracking Method
  - When you begin to perform analysis, this rule will make more sense than it probably does now but it is important to keep in mind.
  - When you manipulate vector spatial data files, you will frequently be generating a copies of the data that have “added value”.
  - When you manipulate raster data files, you will frequently be performing a process multiple times before you get the outcome you wish.



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- Therefore, it is important that you have a naming convention that will allow you to keep track of these sequential files. Example, you may start with a shapefile of town\_boundaries which only has one attribute (e.g. the name of the town). Performing a process called a Join, results in new attributes (population of each town, zip code, the number of grocery stores in each town, etc) being appended to the table. Once the join is completed, you would export the entire file to a new file in order to permanently retain the new attributes. Because you have implemented a tracking method it will be easy to distinguish the latest file you are working with.



## Revision Questions

**EXERCISE 1.**  Define georeferencing.

**Example**  . Distinguish between vector and raster.

*Solution:* Revise.

**EXERCISE 2.**  List three rules to follow in creating data files.



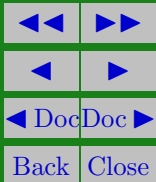
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## Solutions to Exercises

### Exercise 1. Revise.

### Exercise 1





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## Exercise 2. Revise.

## Exercise 2

